



Optimizing UV Stabilization in PVC **using HALS**

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Abstract

Polyvinyl Chloride (PVC) degrades under UV radiation, posing a challenge for its outdoor use [1]. This study develops a kinetic model to predict PVC's degradation rate and evaluate the protective effect of the Hindered Amine Light Stabilizer (HALS) Tinuvin 770. By simulating the chemical reactions involved in PVC degradation and the stabilizing action of HALS [2] [3], the model estimates PVC's lifespan under UV exposure. Results show that Tinuvin 770 effectively slows down PVC degradation, extending the material's usable lifetime. A user-friendly tool is also developed to predict the optimal stabilizer concentration, aiding in the design of PVC formulations with enhanced durability. This research provides a practical method for improving the longevity and performance of PVC in outdoor applications, contributing to the development of more sustainable and cost-effective solutions for industries reliant on this versatile polymer.

1. Introduction

Polyvinyl Chloride (PVC), a versatile thermoplastic polymer known for its durability, chemical resistance, and cost-effectiveness, is widely used in various sectors, from construction and infrastructure to healthcare and consumer goods. The global PVC market has shown robust growth, driven by its wide-ranging applications and favorable properties. Polyvinyl Chloride (PVC) Market size was valued at USD 82.52 Billion in 2023 and is projected to reach USD 110 Billion By 2030, growing at a CAGR (Compound Annual Growth Rate) of 2.95% during the forecast period 2024 to 2030.



Graph 01: Global PVC market growth [2].

PVC chemical structure, characterized by repeating vinyl chloride units, contributes to its desirable properties. However, PVC is susceptible to degradation when exposed to ultraviolet (UV) radiation, particularly in outdoor applications where prolonged sunlight exposure is inevitable. This

degradation can lead to discoloration, embrittlement, and loss of mechanical strength, significantly compromising its performance and lifespan [4].

The degradation of PVC under UV radiation involves a complex series of chemical reactions initiated by the absorption of UV photons. This process often generates free radicals, such as chlorine radicals and polyenyl radicals, which propagate the degradation further. A major consequence of PVC degradation is the release of hydrochloric acid (HCl), contributing not only to the polymer's deterioration but also raising environmental concerns due to its corrosive nature [5].

To mitigate UV-induced degradation, various stabilization strategies have been employed, including the use of UV absorbers, antioxidants, and Hindered Amine Light Stabilizers (HALS). Among these, HALS, particularly Tinuvin 770, have proven to be effective due to their ability to scavenge free radicals and inhibit the chain reactions responsible for PVC degradation. Despite the known effectiveness of HALS, determining the optimal concentration of Tinuvin 770 for a specific application and predicting the long-term stability of PVC under varying UV exposure conditions remain critical challenges [6].

This project addresses these challenges by developing a comprehensive kinetic model that accurately predicts the degradation rate of PVC under UV radiation and evaluates the effectiveness of Tinuvin 770 in retarding this process. The model is built upon a detailed understanding of the underlying chemical mechanisms involved in PVC degradation and the stabilizing action of HALS. By formulating and solving a set of differential equations that describe the concentration changes of key species over time, valuable insights into the kinetics of the degradation process and the influence of parameters such as UV intensity, temperature, and stabilizer concentration can be obtained.

The development of this kinetic model has far-reaching implications for industries utilizing PVC in outdoor applications. It offers a quantitative tool for predicting the long-term performance and durability of PVC products, enabling manufacturers to optimize the formulation of PVC compounds and select appropriate stabilizer concentrations to ensure adequate protection against UV degradation. This project is crucial to the global PVC market, which holds significant economic importance due to its extensive applications in construction, automotive, healthcare, and consumer goods sectors. Ensuring the durability and performance of PVC in outdoor environments directly impacts the industry's market share and competitive edge. By developing an advanced kinetic model to predict UV-induced degradation and evaluate the efficacy of stabilizers, this research provides manufacturers with the tools needed to enhance product longevity and reliability. This capability is vital for sustaining customer trust and reducing costs associated with material failure and replacements. The insights gained from this project will support more sustainable practices, further strengthening the market position of PVC by addressing environmental concerns and regulatory requirements. As a result, this project not only aids in improving product quality but also contributes to the overall growth and sustainability of the global PVC market. Moreover, the model can be used to assess the potential environmental impact of PVC degradation by quantifying HCl release under different exposure scenarios, contributing to more sustainable practices in the use of this ubiquitous polymer.

2. Project Objectives

The primary objectives of this project are to:

1. Develop a comprehensive kinetic model that simulates the degradation of PVC under UV exposure, incorporating the relevant chemical reactions and rate equations.
2. Integrate the action of HALS (Tinuvin 770) into the model, considering its interaction with radical species and the resulting stabilization mechanisms.
3. Create a user-friendly software interface that enables users to input parameters such as PVC formulation, UV intensity, and desired product lifetime, and then predicts the optimal concentration of Tinuvin 770 required to achieve the desired UV resistance.

3. Photodegradation

Long term exposure to sunlight leads to the degradation of plastic materials. UV energy absorbed by plastics can excite photons, which then create free radicals. While many pure plastics cannot absorb UV radiation, the presence of catalyst residues and other impurities will often act as free radical receptors, and degradation occurs [7]. It only takes a very small amount of impurity for the degradation to occur. In the presence of oxygen, the free radicals form oxygen hydroperoxides that can break the double bonds of the backbone chain leading to a brittle structure. This process is often called photo-oxidation. However, in the absence of oxygen there will still be degradation due to the cross-linking process [8]. Poly vinyl chloride has poor light stability in the wavelength range of 253–310 nm, presumably due to the presence of unsaturated (C=C) bonds, carbonyl, hydroperoxide, and hydroxyl groups in polymer chains. The relative activity of one or another chromophore in initiating PVC photodegradation is determined primarily by two factors: their ability to absorb UV light in the wavelength range under consideration and their participation in the formation of active particles (radicals) which cause degradation of the polymer chains. The alkene unsaturated (C=C) bonds (both internal and terminal) cannot be the primary initiators of PVC photodegradation under the action of sunlight with ($\lambda > 250$) nm because they absorb only the UV light at ($\lambda < 200$) nm. The absorption by conjugated (C=C) bonds (dienes, trienes, etc.) shifts toward the longer wave lengths [7]. The color changes from white to yellow, brown, and finally to black while the properties of the material deteriorate.

3.1 Mechanisms of Photodegradation

UV Absorption

PVC is particularly susceptible to photodegradation because it absorbs UV radiation, especially in the UV-B range (280-320 nm). When PVC absorbs UV radiation, the energy from the UV light excites the electrons in the polymer chains. This excitation elevates the energy levels of the electrons, making the bonds within the PVC molecules more unstable and prone to breaking. The primary regions within the PVC molecule that absorb this UV radiation are the C-Cl (carbon-chlorine) bonds and certain unsaturated sites within the polymer [9].

Free Radical Formation

The breakage of bonds during chain scission generates highly reactive species known as free radicals. Free radicals are atoms or molecules that contain unpaired electrons, making them extremely reactive [10]. In the context of PVC degradation, these free radicals can react with other polymer chains or with oxygen in the air, initiating further degradation reactions. For example, the free radicals can react with oxygen to form peroxy radicals, which can further decompose to produce additional free radicals, perpetuating a cycle of degradation [9] [10].

Chain Scission and Cross-Linking

PVC is particularly susceptible to photodegradation because it absorbs UV radiation, especially in the UV-B range (280-320 nm). When PVC absorbs UV radiation, the energy from the UV light excites the electrons in the polymer chains. This excitation elevates the energy levels of the electrons, making the bonds within the PVC molecules more unstable and prone to breaking. The primary regions within the PVC molecule that absorb this UV radiation are the C-Cl (carbon-chlorine) bonds and certain unsaturated sites within the polymer [11] [12].

3.1.1 Photooxidation

3.1.1.1 Oxygen Interaction

During UV exposure, the energy absorbed by PVC leads to the formation of free radicals as the polymer chains undergo bond cleavage [12]. These free radicals are highly reactive and can interact with oxygen present in the environment, leading to a process known as photooxidation. When free radicals react with oxygen, they form peroxides and hydroperoxides [12]. These oxidative species are unstable and can decompose to generate more free radicals, perpetuating a cycle of oxidative degradation. The presence of oxygen thus significantly accelerates the degradation process, as the formation and decomposition of peroxides and hydroperoxides contribute to the continual breakdown of the polymer structure.

3.1.1.2 Oxidative Chain scission

The oxidative products formed during photooxidation, such as peroxides and hydroperoxides, can further react with the polymer chains, leading to chain scission and cross-linking. Oxidative chain scission occurs when these peroxides and hydroperoxides decompose, generating additional free radicals that break the polymer chains into smaller segments. This results in a decrease in molecular weight and a deterioration of mechanical properties, such as tensile strength and flexibility. Additionally, some of the free radicals can induce cross-linking, where the polymer chains become interconnected in a network structure [12]. While cross-linking can sometimes enhance certain properties like rigidity, it generally leads to increased brittleness and reduced flexibility in PVC. Overall, the oxidative degradation severely compromises the material's mechanical integrity and performance.

3.1.2 Cross-linking

Cross-linking is a significant secondary reaction that occurs when free radicals generated during the degradation of PVC interact with adjacent polymer chains [12]. These free radicals can initiate reactions between different polymer chains, causing them to bond together, forming a three-dimensional network structure. This cross-linking process alters the material properties of PVC, typically making it more rigid and less flexible [13]. As the polymer chains become interconnected, the material loses its ability to deform easily under stress, leading to increased brittleness. While cross-linking can sometimes provide enhanced structural stability, in the case of PVC degradation, it usually results in undesirable mechanical properties. The material becomes more prone to cracking and failure

under mechanical loads, thus compromising its performance and reducing its lifespan in applications where flexibility and toughness are essential [13] [14] [15]

3.1.4 Mechanistic Pathways

Primary Photochemical Processes: In the presence of UV radiation, excited singlet polyenes form and undergo various degradation pathways, resulting in a range of degradation products and structural changes.

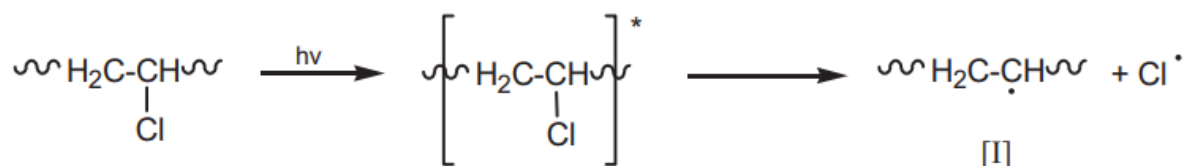


Figure 01: Formation of polyene radical [9] [14] [15]

Free Radical Reactions: Polyenes and other intermediates react with oxygen, forming peroxy radicals and other degradation products, leading to chain scission and cross-linking.

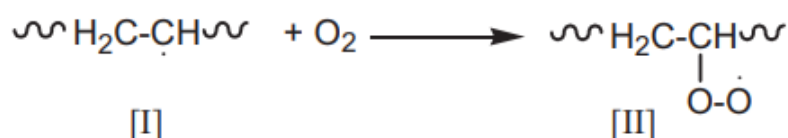


Figure 02: Formation of peroxy radical [9] [14] [15]

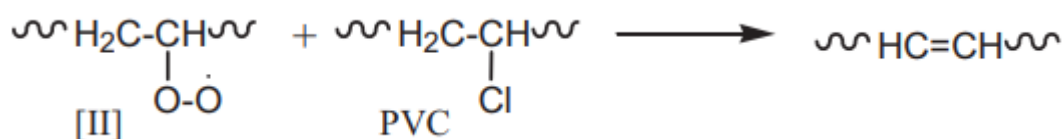


Figure 03: Formation of PVC radical [9] [14] [15]

Understanding these mechanisms is crucial for developing the overall photodegradation mechanism of PVC

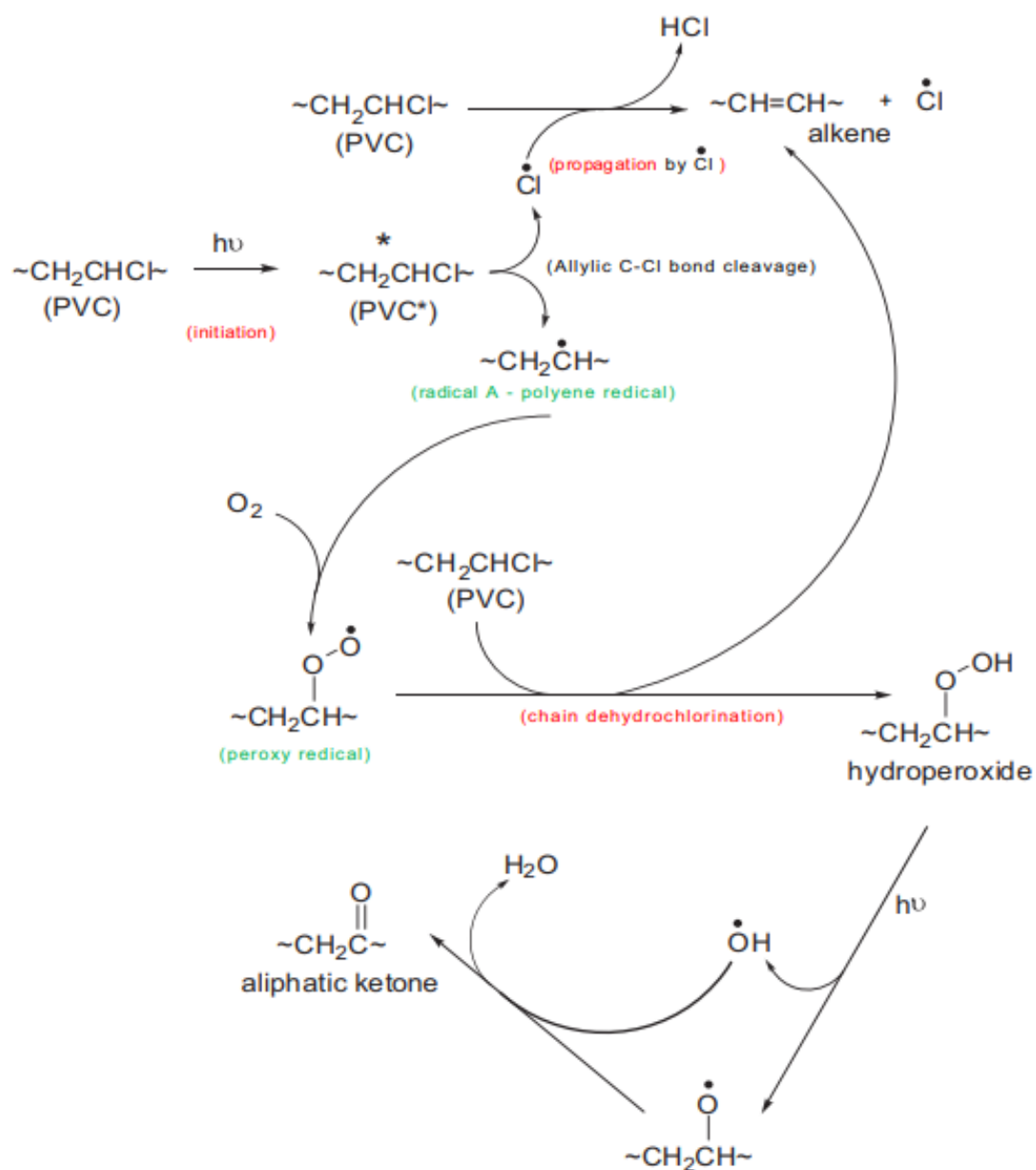


Figure 04: Overall photodegradation mechanism [9] [14] [15]

3.2 Effects of UV Degradation on PVC Properties

1. Discoloration

Yellowing

One of the most noticeable effects of UV degradation on PVC is yellowing, primarily due to the formation of conjugated polyenes. This occurs through a process called dehydrochlorination, where UV radiation breaks the carbon-chlorine bonds in the PVC polymer, releasing hydrogen chloride (HCl) gas. The removal of HCl from the polymer chain leaves behind unsaturated double bonds, which can align to form conjugated systems known as polyenes. These polyenes absorb light in the visible spectrum, particularly in the blue region, causing the material to appear yellow. This discoloration not only affects the aesthetic appearance of PVC products but also indicates underlying chemical changes that may compromise the material's performance and longevity [16].



Figure 05: UV Degraded PVC Pipes

Surface Cracking

Prolonged UV exposure can lead to surface cracking and embrittlement of PVC. As the polymer undergoes photodegradation, the repeated breaking of bonds and the formation of free radicals weaken the material's structure. Over time, this degradation leads to the formation of micro-cracks on the surface, which can grow and coalesce into larger cracks. These surface cracks are not merely cosmetic; they indicate a significant loss of mechanical integrity. The cracks can propagate under mechanical stress, leading to failure in applications where the material needs to maintain its strength and flexibility. Surface cracking also increases the material's susceptibility to environmental factors such as moisture and chemical attack, further accelerating the degradation process.

2. Loss of Mechanical Properties

Tensile Strength

UV degradation significantly reduces the tensile strength and elongation at break of PVC. The breaking of polymer chains (chain scission) and the loss of molecular weight weaken the material, making it less capable of withstanding tensile forces. This degradation leads to a reduction in the PVC's ability to stretch and absorb energy before breaking, making it more prone to cracking and

breaking under stress. In applications where PVC is expected to endure mechanical loads, such as pipes, siding, and outdoor furniture, this loss of tensile strength can lead to premature failure and reduced service life. The material's overall durability is compromised, necessitating more frequent repairs or replacements.

Impact Resistance

The impact resistance of PVC also decreases as it becomes more brittle due to UV degradation. The formation of free radicals and subsequent chain scission reduce the polymer's flexibility and toughness. As a result, PVC becomes more brittle and less capable of absorbing and dissipating the energy from impacts. This increased brittleness means that the material is more likely to shatter or crack upon impact, posing a risk in applications where resistance to sudden forces is crucial. For example, PVC used in construction materials, automotive components, or protective equipment may fail to provide the necessary protection or durability, leading to potential safety hazards and increased maintenance costs.

3. Surface Roughness

Texture Changes

UV exposure can cause significant changes in the surface texture of PVC, leading to increased roughness and the development of micro-cracking. As the polymer degrades, the surface undergoes a series of chemical and physical transformations. The loss of plasticizers, additives, and the breakdown of polymer chains contribute to the roughening of the surface. Micro-cracks begin to form as the material becomes more brittle, creating a rough and uneven texture [16]. This increased surface roughness can have several negative implications, including a higher likelihood of dirt accumulation, microbial growth, and reduced aesthetic appeal. In applications where a smooth surface is essential for performance, such as in coatings, films, or medical devices, these texture changes can significantly impair functionality and necessitate additional surface treatments or protective coatings.

3.3. Role of Stabilizers in Mitigating Degradation

3.3.1 Hindered Amine Light Stabilizers (HALS)

Hindered amine light stabilizers (HALS) have been well recognized for more than 25 years to be the most proficient UV stabilizers for a large number of polymers. This is particularly true for the protection of polyolefins HALS against the degradative effects of light energy [17].

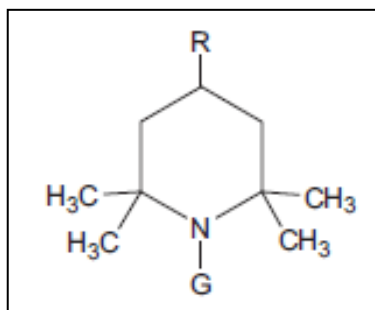


Figure 06: HALS ring Structure [3]

Free Radical Scavenging

Hindered Amine Light Stabilizers (HALS) are highly effective in mitigating the degradation of PVC by scavenging free radicals formed during the photodegradation process. Free radicals, generated when PVC absorbs UV radiation, can initiate a series of degradation reactions, including chain scission and cross-linking, which compromise the material's mechanical properties. HALS neutralize these free radicals by stabilizing them and preventing them from reacting further. This scavenging action interrupts the degradation cycle, thereby protecting the polymer chains from breaking down. By preventing the propagation of free radicals, HALS help maintain the integrity and longevity of PVC products exposed to UV radiation [3].

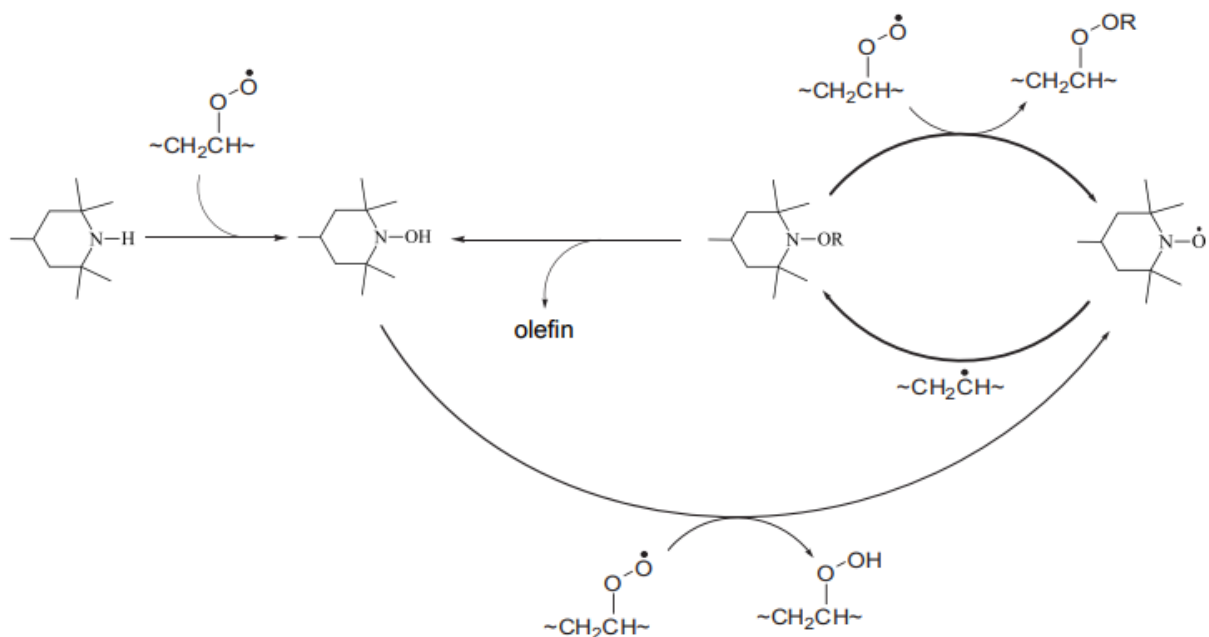


Figure 07: Reaction and regeneration mechanism of HALS [3]

Regeneration

One of the unique advantages of HALS is their regenerative capability. Unlike other stabilizers that may become depleted over time, HALS can regenerate after reacting with free radicals. This means they can provide long-term protection for PVC, even under prolonged UV exposure. The regenerative nature of HALS ensures a sustained defense against photodegradation, making them particularly valuable for applications where long-lasting UV stability is crucial. This ongoing protection helps preserve the mechanical strength, appearance, and overall performance of PVC products, reducing the need for frequent replacements or maintenance.

3.3.2 UV Absorbers

Absorbers convert harmful ultraviolet radiation to harmless infrared radiation or thermal energy, which is dissipated through the polymer matrix.

Energy Dissipation

UV absorbers, such as benzophenones and benzotriazoles, play a crucial role in protecting PVC by absorbing harmful UV radiation and dissipating it as harmless thermal energy. These stabilizers are designed to absorb UV light within specific wavelength ranges, preventing the energy from reaching and damaging the polymer chains. By converting the absorbed UV energy into heat, which is then harmlessly dissipated, UV absorbers effectively shield the PVC material from the harmful effects of UV radiation. This process helps maintain the structural integrity and aesthetic quality of PVC, making UV absorbers essential for applications exposed to sunlight.

Surface Protection

UV absorbers are particularly effective in protecting the surface layers of PVC from UV radiation. Since UV light primarily affects the material's surface, incorporating UV absorbers into the top layers of PVC products provides an effective barrier against photodegradation. This surface protection is critical in maintaining the appearance and durability of PVC products, such as outdoor furniture, siding, and automotive components. By preventing surface degradation, UV absorbers help ensure that PVC products retain their color, gloss, and mechanical properties over time, enhancing their performance and extending their service life.

3.3.3 Antioxidants

Oxidation Prevention

Antioxidants are often used in conjunction with UV stabilizers to provide comprehensive protection against oxidative degradation. During photooxidation, free radicals can react with oxygen to form peroxides and hydroperoxides, which further degrade the polymer. Antioxidants work by neutralizing these oxidative species, preventing them from causing additional damage. By intercepting and deactivating peroxides and hydroperoxides, antioxidants inhibit the oxidative chain reactions that lead to the breakdown of PVC. This dual approach, combining UV stabilizers and antioxidants, offers robust protection against both photodegradation and oxidation, ensuring the long-term stability and performance of PVC products [13].

3.4 Experimental Data on PVC Degradation

To fully understand the degradation mechanisms and the effectiveness of stabilizers, experimental data is crucial. Here are some typical experimental approaches:

3.4.1 Accelerated Weathering Tests

UV Exposure Chambers

The UV accelerated weathering test chamber works by subjecting the material to intense UV radiation, which is the most damaging component of sunlight, along with controlled temperature and humidity conditions.

The chamber consists of a large, enclosed space with UV lamps installed on the walls or ceiling. These lamps emit high-intensity UV radiation, mainly in the UVA and UVB range, which closely replicates the solar spectrum. The chamber also incorporates a system to control temperature and humidity levels, allowing for the simulation of different weather conditions.

During the test, the material to be evaluated is placed inside the chamber and exposed to the UV radiation for a specified duration. The temperature and humidity conditions can be varied according to the desired test parameters, such as high temperature and high humidity for tropical climates or low temperature and low humidity for arid climates [18].



Figure 08: Accelerated Weathering Test Chamber [18]

The UV radiation causes the material to undergo various degradation processes, such as photochemical reactions, oxidation, and polymer chain scission. These reactions can lead to changes in the material's physical, chemical, and mechanical properties, including color fading, surface cracking, loss of strength, and degradation of surface coatings.

To evaluate the weather resistance of the material, samples are periodically removed from the chamber at specific time intervals and subjected to various tests and inspections. These tests may include visual examination, color measurement, mechanical testing, chemical analysis, and microscopic analysis.

By exposing the material to accelerated weathering conditions, the UV accelerated weathering test chamber allows for the prediction of long-term performance in a relatively short period of time. The results obtained from these tests help manufacturers, researchers, and quality control personnel to assess the durability, lifespan, and suitability of materials for outdoor applications, such as building materials, automotive components, coatings, and textiles.

Measurement of Property Changes

During and after exposure in UV chambers, various properties of PVC are meticulously measured to assess degradation. Mechanical properties such as tensile strength, elongation at break, and impact resistance are tested to determine how UV exposure affects the material's durability. Additionally, changes in color are monitored to evaluate discoloration and yellowing, while surface texture analyses reveal the extent of surface cracking and roughness. These measurements provide a comprehensive understanding of how UV radiation deteriorates PVC, highlighting the specific areas where stabilizers can improve performance. This data is crucial for developing and optimizing stabilizers that enhance the UV resistance of PVC.

3.4.2 Spectroscopic Analysis

FTIR (Fourier Transform Infrared Spectroscopy)

Fourier Transform Infrared Spectroscopy (FTIR) is instrumental in studying PVC degradation by identifying chemical changes and detecting degradation products. FTIR operates by exposing the PVC sample to infrared light and measuring the absorption spectrum. Each chemical bond within the PVC molecule absorbs infrared radiation at unique frequencies, generating characteristic peaks in the FTIR spectrum. For PVC, FTIR can detect changes indicative of degradation, such as the appearance of carbonyl (C=O) stretching vibrations due to oxidation processes. This allows researchers to monitor the degradation of PVC under different environmental conditions, including exposure to UV radiation or thermal stress. By comparing FTIR spectra before and after exposure, one can quantify the extent of degradation and assess the effectiveness of stabilizers and other additives in preserving PVC's integrity and performance [19].

Initially, original tensile strength, percentage elongation, hardness (Shore D) and water absorption properties were measured. After that prepared samples were subjected to the UV exposure test under the standard conditions. Seven set of samples were removed from the UV chamber at different time intervals of 24, 48, 72, 96, 120, 144 and 168 hours. Tensile strength, percentage elongation, hardness, water absorption properties and visual inspection tests were done after the UV exposure treatment.

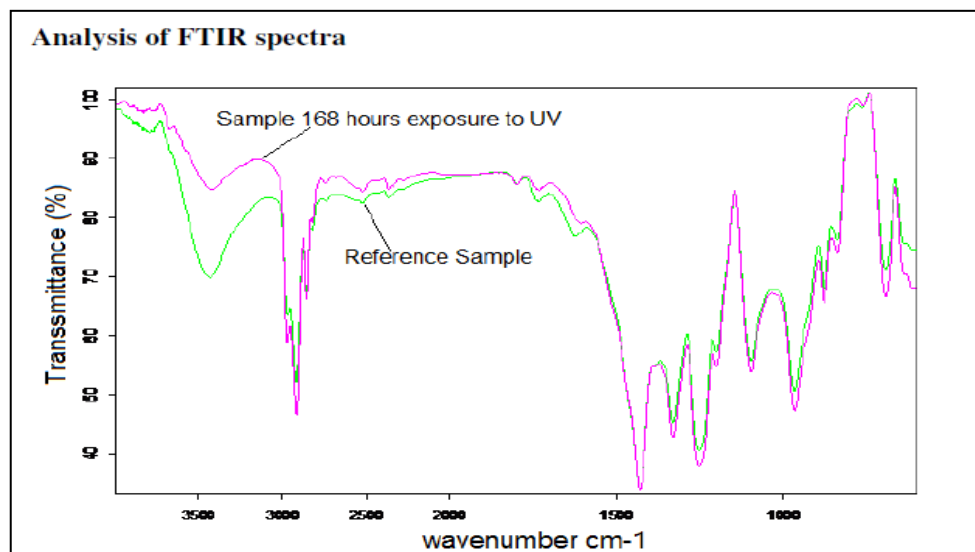
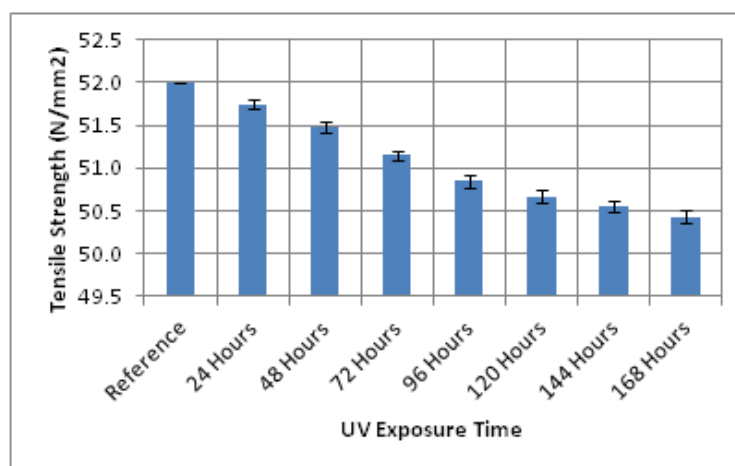


Figure 08: FTIR spectrums of reference sample and samples exposed to UV up to 168 hours. [19]

3.5 Analysis of Mechanical Properties

1. Tensile Strength

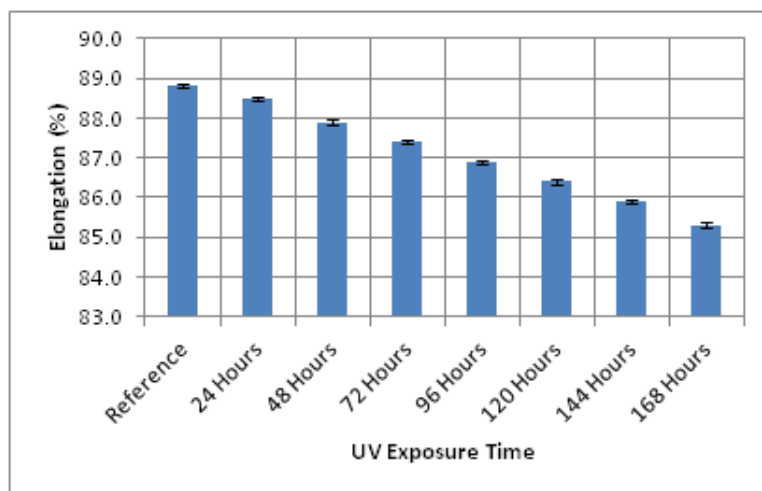
Tensile test results in Figure, showed the gradual reduction of tensile strength with increase of UV exposure time. Gradual reduction of tensile strength of samples may be due to the reduction of molecular-chain stiffness, caused by chain scissions due to UV attack.



Graph 02: Tensile strength of reference sample and variation with exposure time UV. [19]

Elongation at Break

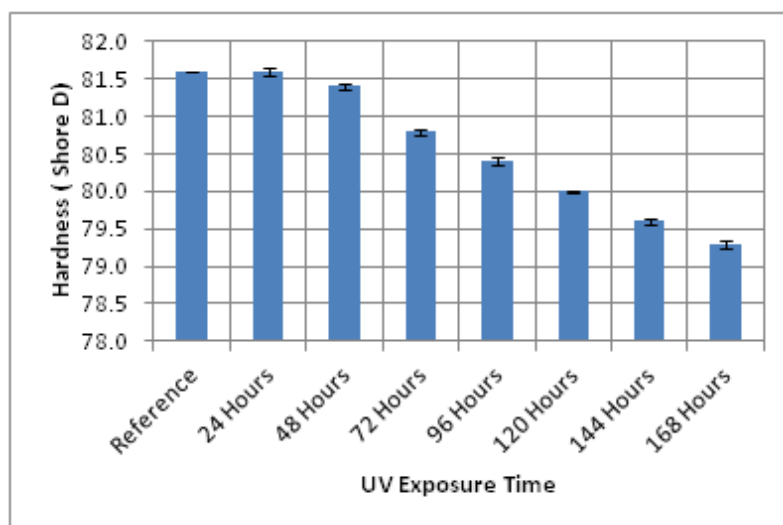
Test results of percentage elongation at break also reduced somewhat gradually and slightly with the increase of UV exposure time (Figure). The causes of reduction of tensile strength could have also attributed for a gradual reduction in the elongation of sample. During the degradation, structure becomes embrittled and molecular weight reduced



Graph 03: Relative elongation of sample with exposure time of UV. [19]

Hardness Properties

According to the Figure, there is a gradual reduction of hardness with increase of UV exposure time. Reduction of hardness may be due to the reduction of structural stiffness or increase of surface softness caused by the degradation process [19].



Graph 04: Hardness of sample with exposure time of UV. [19]

UV-Vis Spectroscopy

UV-Vis spectroscopy is a technique used to study how PVC interacts with UV and visible light. PVC can degrade when exposed to UV radiation, which affects its color, strength, and overall durability. UV-Vis spectroscopy measures the absorbance of light by PVC across different wavelengths, revealing which wavelengths PVC absorbs and how much. This helps in understanding how UV radiation affects PVC and how additives like UV absorbers or stabilizers can protect it. UV absorbers are chemicals added to PVC formulations to absorb UV light and prevent it from damaging the polymer chains. By analyzing the absorbance spectrum of PVC samples with and without UV absorbers, researchers can determine the effectiveness of these additives in shielding PVC from UV-induced degradation [20].

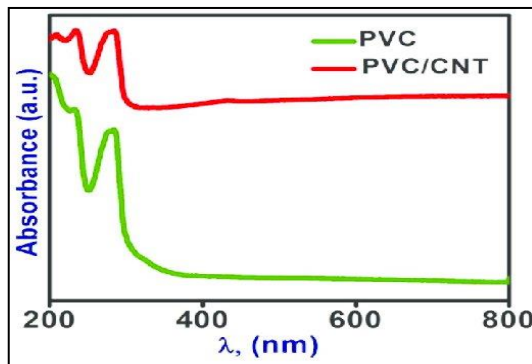


Figure 09: Comparison [20]

UV-Vis spectroscopy also helps identify the optimal wavelengths for UV absorption by PVC and UV absorbers. This information guides the selection and formulation of PVC products with enhanced UV resistance. By studying absorbance changes before and after UV exposure, manufacturers can tailor PVC formulations to withstand environmental conditions and extend product lifespan. Overall, UV-Vis spectroscopy provides critical insights into how PVC interacts with light, aiding in the development of more durable and reliable PVC materials for various applications, from construction to consumer goods, where UV stability is essential [21].

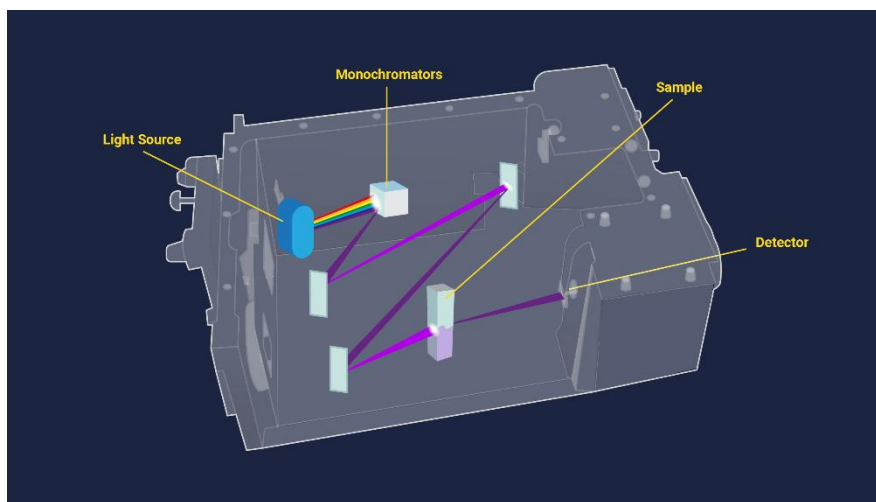


Figure 10: Schematic diagram of a cuvette-based UV-Vis spectroscopy system [21]

1. Thermal Analysis

TGA (Thermogravimetric Analysis)

Thermogravimetric analysis (TGA) is a powerful technique used to assess the thermal stability and decomposition kinetics of PVC (Polyvinyl Chloride). During TGA, a sample of PVC is heated at a controlled rate in an inert atmosphere while its weight change is continuously monitored. This method allows researchers to observe how PVC behaves as it is heated, revealing important information about its thermal stability. PVC typically undergoes degradation processes such as the release of hydrogen chloride gas and the breakdown of its polymer chains at specific temperatures. TGA provides precise data on the onset, rate, and extent of these degradation processes, which are crucial for determining the thermal limits of PVC in various applications. By analyzing TGA curves, researchers can also derive kinetic parameters such as activation energy, which quantifies the energy required for PVC degradation reactions. This data helps in predicting the service life and performance of PVC under different thermal conditions and informs decisions on material selection and formulation [22].

DSC (Differential Scanning Calorimetry)

Differential scanning calorimetry (DSC) complements TGA by providing detailed insights into the thermal properties of PVC before and after degradation. DSC measures the heat flow into or out of a sample as it is heated or cooled, revealing phase transitions, melting points, and changes in specific heat capacity. For PVC, DSC is particularly useful in detecting changes in these thermal properties due to degradation processes such as chain scission, cross-linking, or crystallization. By analyzing DSC thermograms, researchers can pinpoint shifts in these thermal events, indicating how PVC's molecular structure and physical properties are altered by thermal stress. This information is critical for assessing the impact of degradation on PVC's mechanical strength, flexibility, and dimensional stability. Overall, DSC provides essential data to understand how thermal degradation affects PVC's performance and guides efforts to enhance its thermal stability through material design and additive selection [23].

4. Kinetic Analysis of Photodegradation

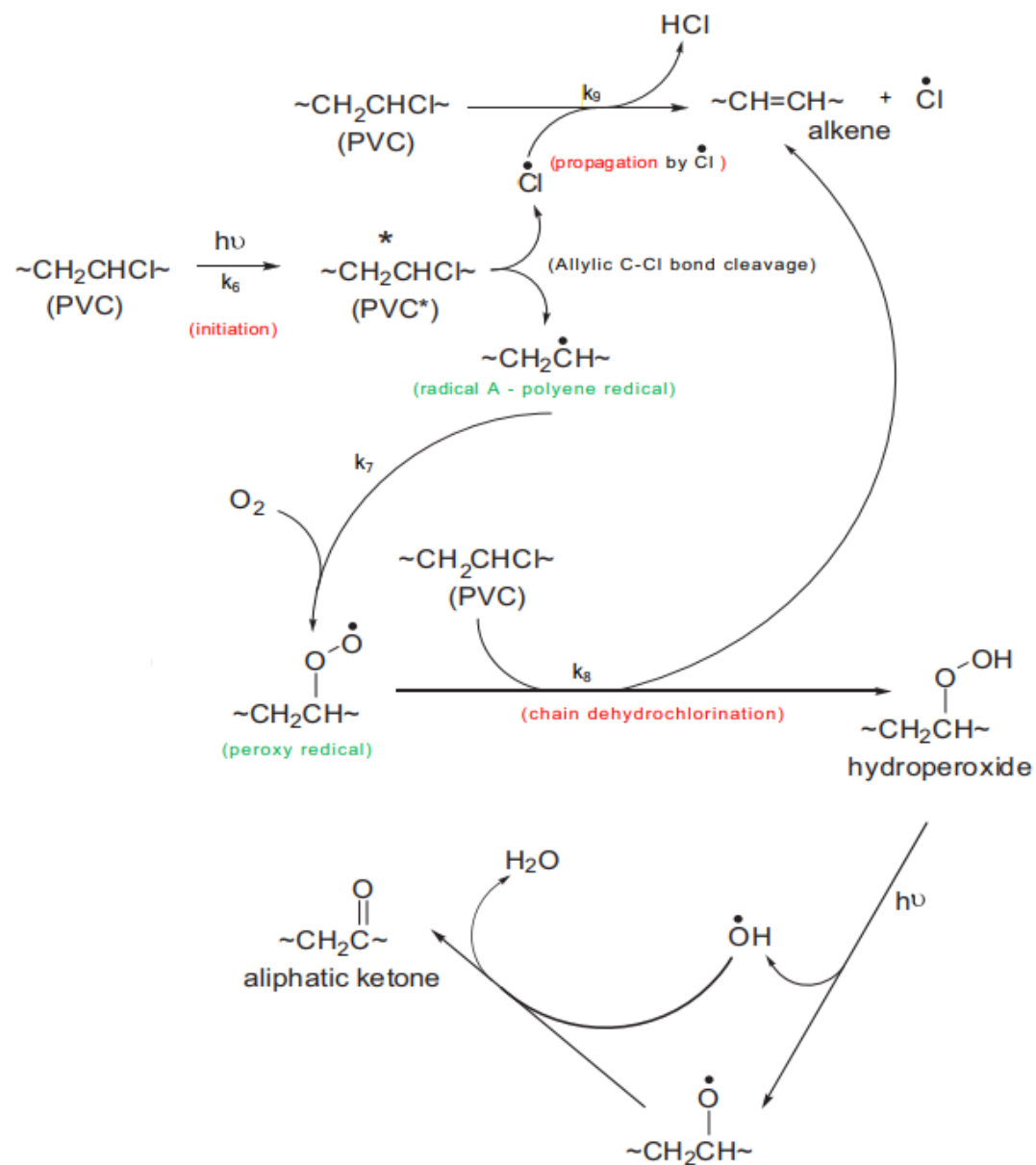


Figure 11: Overall photodegradation mechanism with rate constants [9] [14] [15]

Without stabilizer,

Concentration of PVC:

$$\frac{d[PVC]}{dt} = -k_6 [PVC] - k_8 [PVC] [R'OO \bullet] - k_9 [Cl \bullet][PVC]$$

Concentration of Alkyl Radicals ($R' \bullet$):

$$\frac{d[R' \bullet]}{dt} = k_6 [PVC] - k_7 [R' \bullet]$$

Concentration of Peroxy Radicals ($R'OO \bullet$):

$$\frac{d[R'OO \bullet]}{dt} = k_7 [R' \bullet] - k_8 [R'OO \bullet][PVC]$$

Concentration of Peroxy Radicals ($Cl \bullet$):

$$\frac{d[Cl \bullet]}{dt} = k_6 [PVC] + k_8 [PVC][R'OO \bullet] - k_{10} [Cl \bullet]^2$$

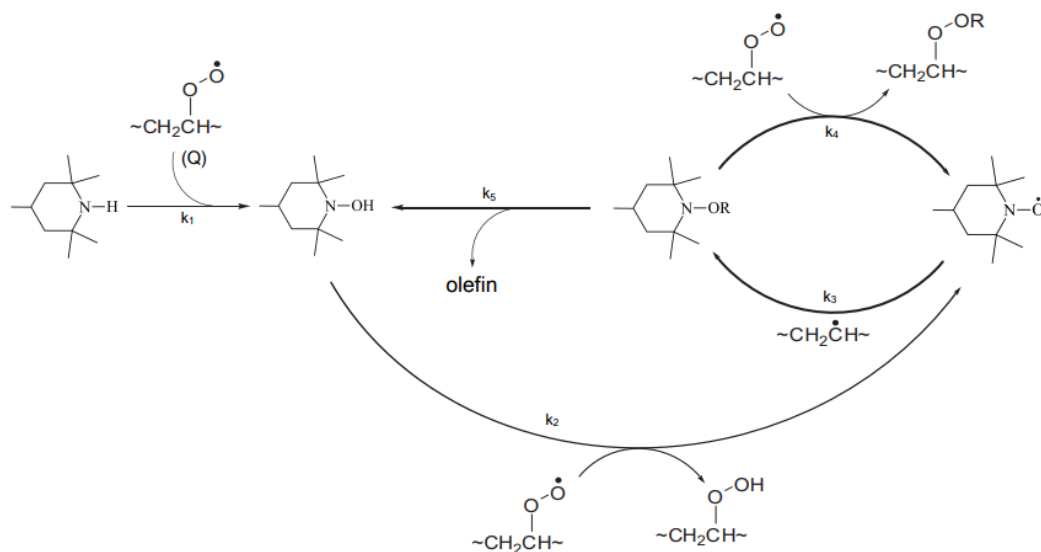


Figure 12: Reaction and regeneration mechanism of HALS with rate constants [3]

With stabilizer,

Concentration of PVC:

$$\frac{d[PVC]}{dt} = -k_6 [PVC] - k_8 [PVC] [R'OO \bullet] - k_9 [Cl \bullet] [PVC]$$

Concentration of Peroxy Radicals (Cl•):

$$\frac{d[Cl \bullet]}{dt} = k_6 [PVC] - k_8 [PVC] [R'OO \bullet]$$

Concentration of N-Hydroxy HALS (R-N-OH):

$$\begin{aligned} \frac{d[R - N - OH]}{dt} &= k_1 [R'OO \bullet] [R - N - H] + k_5 [R - N(O - R')] - k_2 [R - N - OH] [R'OO \bullet] \end{aligned}$$

Concentration of Alkoxyamine (R-N(O-R')) :

$$\frac{d[R - N(O - R')]}{dt} = k_3 [R - NO \bullet] [R' \bullet] - k_5 [R - N(O - R')] - k_4 [R - N(O - R')] [R'OO \bullet]$$

Concentration of Nitroxyl Radical (R-NO•):

$$\begin{aligned} \frac{d[R - NO \bullet]}{dt} &= k_4 [R - N(O - R')] [R'OO \bullet] + k_2 [R'OO \bullet] [R - N - OH] - k_3 [R' \bullet] [R - NO \bullet] \end{aligned}$$

Concentration of Alkyl Radicals (R'•):

$$\frac{d[R' \bullet]}{dt} = k_6 [PVC] - k_7 [R' \bullet] - k_3 [R' \bullet] [R - NO \bullet]$$

Concentration of Peroxy Radicals (R'OO•):

$$\begin{aligned} \frac{d[R'OO \bullet]}{dt} &= k_7 [R' \bullet] - k_8 [R'OO \bullet] [PVC] - k_1 [R'OO \bullet] [R - N - H] \\ &\quad - k_2 [R'OO \bullet] [R - N - OH] - k_4 [R - N(O - R')] [R'OO \bullet] \end{aligned}$$

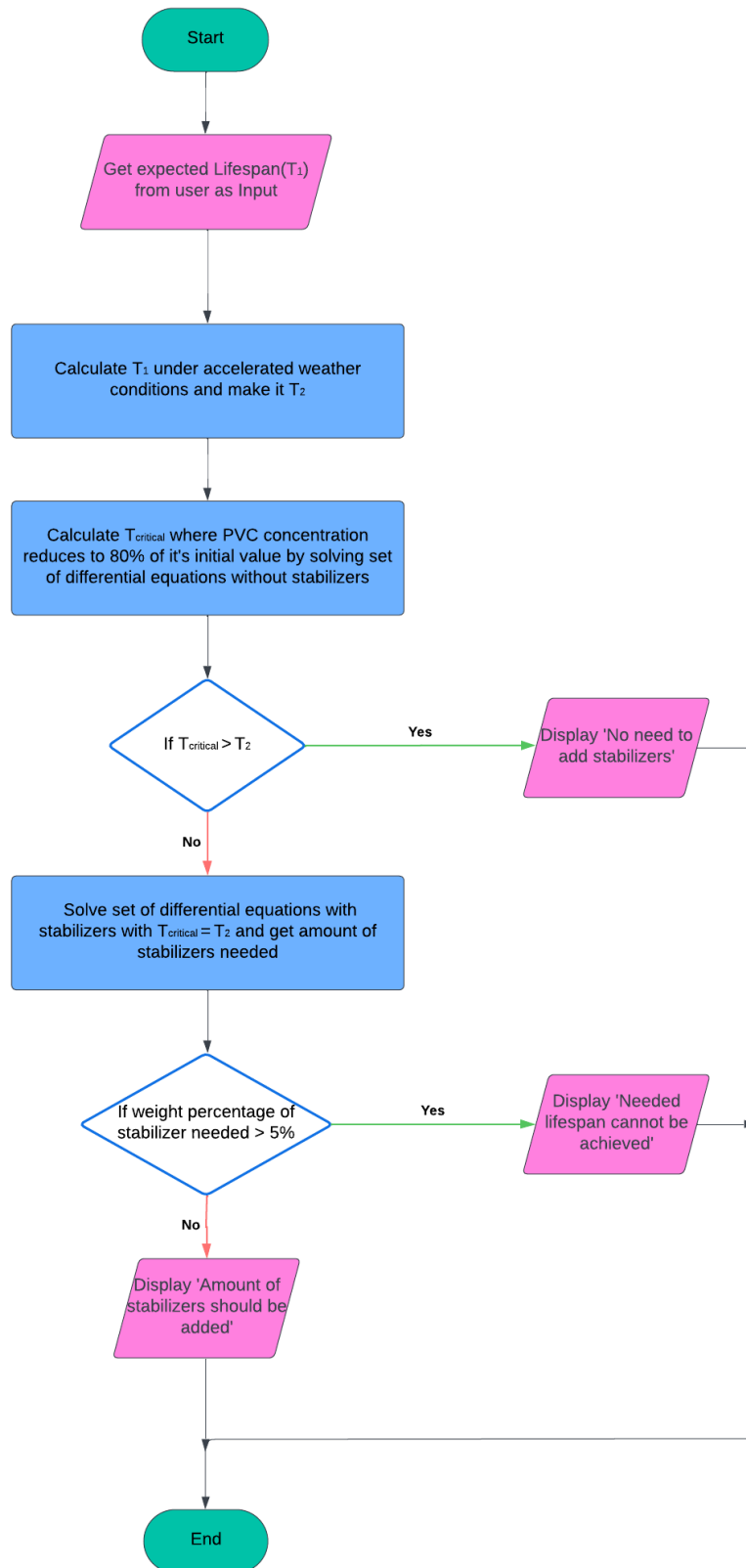
The rate constants of the mechanism were determined under accelerated weather conditions, particularly at a temperature of 50 degrees Celsius and under a UV radiation intensity of 0.76 W/m². These experiments were conducted following the Standard UV testing method: ASTM G154, "Standard Practice for Operating Fluorescent Ultraviolet (UV) Lamp Apparatus for Exposure of Nonmetallic Materials."

Rate Constants	Reference
$k_1 = 51 \text{ M}^{-1} \text{ s}^{-1}$	[9]
$k_2 = 5.5 \times 10^2 \text{ M}^{-1} \text{ s}^{-1}$	[9]
$k_3 = 1.2 \times 10^9 \text{ M}^{-1} \text{ s}^{-1}$	[9]
$k_4 = 1.9 \times 10^{-3} \text{ M}^{-1} \text{ s}^{-1}$	[9]
$k_5 = 9.3 \times 10^{-5} \text{ s}^{-1}$	[24]
$k_6 = 9 \times 10^{-7} \text{ s}^{-1}$	[24]
$k_7 = 1 \times 10^8 \text{ M}^{-1} \text{ s}^{-1}$	[24]
$k_8 = 0.5 \text{ M}^{-1} \text{ s}^{-1}$	[24]
$k_9 = 0.1 \text{ M}^{-1} \text{ s}^{-1}$	[24]

Table 1: Rate constants of mechanism

5. Modeling

5.1 Predictive Algorithm



Input expected lifespan (T_x):

The user enters the desired lifespan of the PVC material under expected weather conditions.

Calculate T:

The algorithm calculates the time it takes for the PVC concentration to degrade to 80% of its initial value under accelerated weather conditions, without any stabilizers. This is achieved by solving a set of differential equations (not shown in the flowchart).

Decision step:

The algorithm compares the calculated T (T_{cal}) to the desired lifespan (T_x).

If T_{cal} is greater than or equal to T_x ($T_{cal} \geq T_x$), then no stabilizers are needed, and the algorithm ends. This indicates that the PVC material will meet the desired lifespan even without stabilizers under normal weather conditions.

Solve with stabilizers:

If T_{cal} is less than T_x ($T_{cal} < T_x$), then stabilizers are required. The algorithm solves another set of differential equations, this time including the presence of stabilizers, to determine the amount of stabilizer needed to achieve the desired lifespan (T_x).

Decision step:

The algorithm checks if the weight percentage of stabilizer needed is greater than 5%.

If the stabilizer percentage is greater than 5%, the algorithm determines that the desired lifespan cannot be achieved with a reasonable amount of stabilizer. This could be due to limitations of the stabilizer or the specific PVC formulation.

Output:

If the stabilizer percentage is less than or equal to 5%, the algorithm outputs the calculated amount of stabilizer required to achieve the desired lifespan (T_x).

This flowchart provides a basic overview of the algorithm for predicting stabilizer content in PVC formulations. The specific details of the differential equations used and the underlying chemical reactions are not shown in the flowchart.

5.2 Calculating Acceleration Factor for Photodegradation of PVC

Step 1: Understand the Variables

- T_{lab} : Laboratory temperature
- I_{lab} : Laboratory light intensity
- t_{lab} : Laboratory exposure time
- T_{env} : Environmental temperature (actual usage condition)
- I_{env} : Environmental light intensity

Step 2: Temperature Acceleration Factor

The Arrhenius equation relates the rate of reaction k to temperature:

$$k = A \exp\left(-\frac{E_a}{RT}\right)$$

Where:

- A is the pre-exponential factor
- E_a is the activation energy
- R is the universal gas constant
- T is the absolute temperature (in Kelvin)

The acceleration factor due to temperature, AF_T , can be calculated as [25]:

$$AF_T = \exp\left(\frac{E_a}{R} \left(\frac{1}{T_{env}} - \frac{1}{T_{lab}}\right)\right)$$

Step 3: Light Intensity Acceleration Factor

The rate of photodegradation is also proportional to light intensity. If we assume a direct proportionality (which is a simplification but often used), the acceleration factor due to light intensity, AF_I , can be calculated as:

$$AF_I = \frac{I_{lab}}{I_{env}}$$

Step 4: Combined Acceleration Factor

The combined acceleration factor, AF , considering both temperature and light intensity, is the product of the individual factors:

$$AF = AF_T \times AF_I$$

Step 5: Calculate Equivalent Environmental Exposure Time

To find the equivalent environmental exposure time t_{env} based on the laboratory exposure time t_{lab} , use:

$$t_{env} = t_{lab} \times AF$$

$$E_a = 230 \text{ kJ mol}^{-1}$$

$$T_{env} = 25^\circ\text{C}$$

$$T_{lab} = 50^\circ\text{C}$$

$$I_{lab} = 0.76 \text{ W m}^{-2}$$

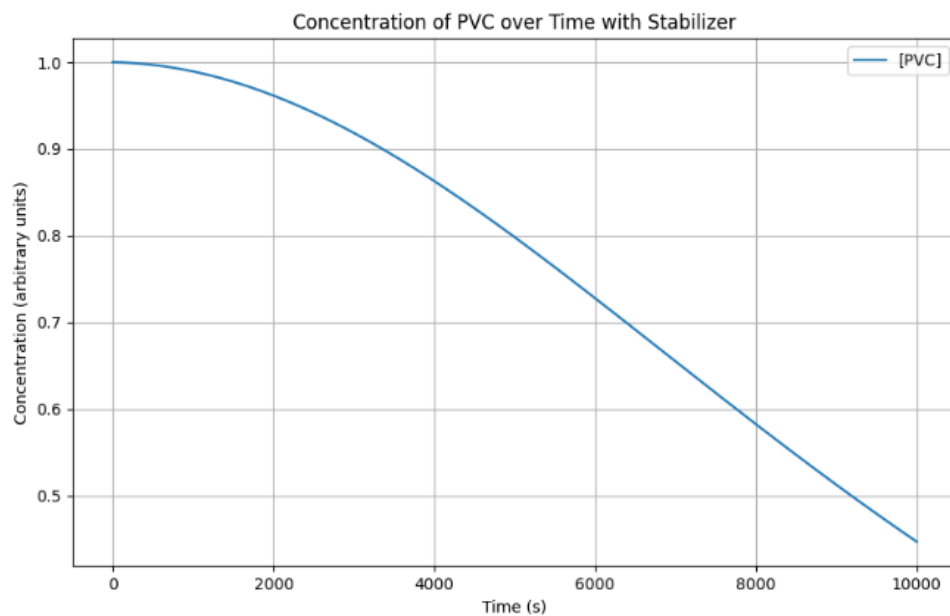
$$I_{env} = 1.33 \text{ W m}^{-2}$$

Therefore, we can calculate AF to be ~ 700

6. Validating Results

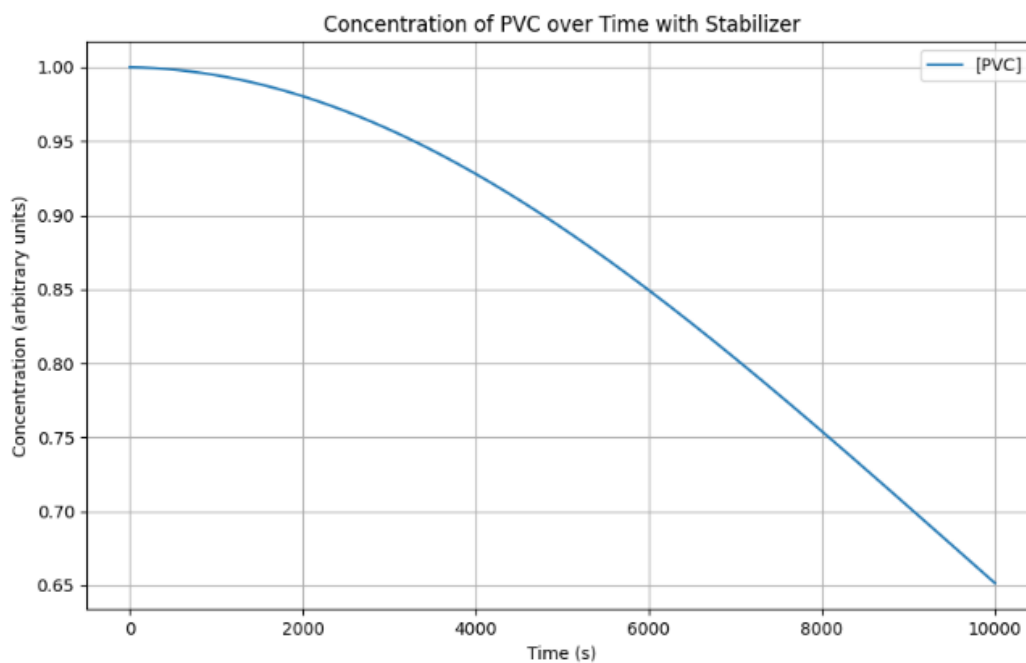
These are the PVC concentration value changing graph plotted without stabilizers and with stabilizers.

- With HALS = 0: This scenario represents a formulation with a negligible amount of HALS, essentially no stabilization effect.



Graph 05: PVC concentration change when stabilizer content is zero

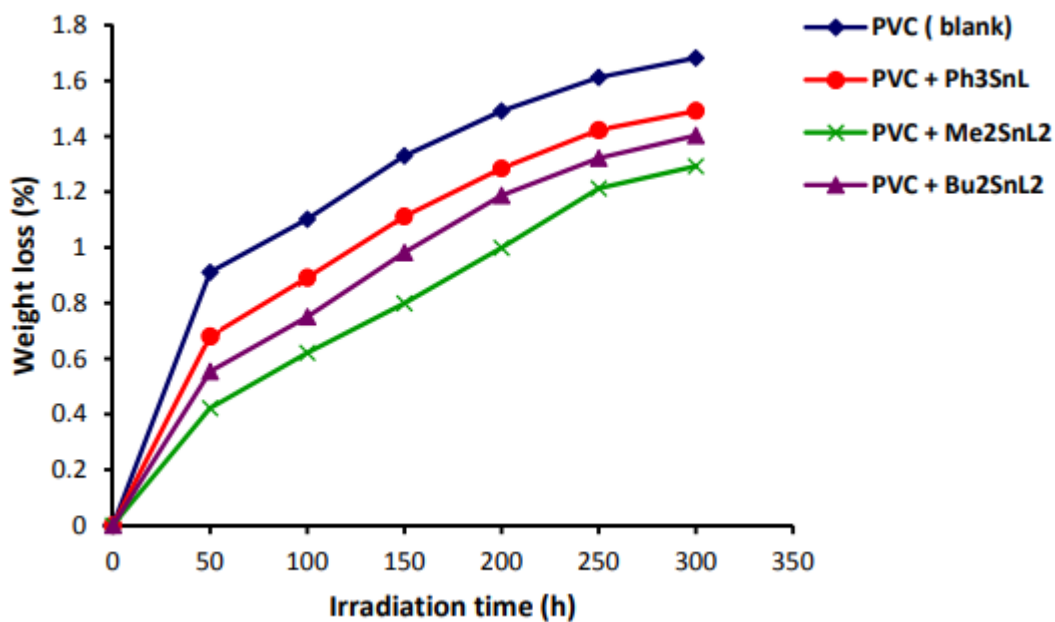
- With HALS = 0.5: This scenario demonstrates the impact of a specific HALS concentration (0.5%wt) on the PVC degradation profile.



Graph 06: PVC concentration change when stabilizer content is 0.5%wt

These plots visually illustrate the effectiveness of HALS in slowing down PVC degradation.

If we correlate these calculated values with the accelerated aging test results, we can reach real-life experimental outcomes. This linkage between predictive modeling and empirical data demonstrates the potential accuracy and reliability of our project in forecasting the real-world performance of PVC materials.



Graph 07: Changes in the weight loss of PVC [26]

7. Proposed Modifications and Enhancements

To further improve the effectiveness of UV stabilizers in protecting PVC from photodegradation, several modifications and enhancements can be considered. These modifications will help in providing a more comprehensive analysis and improving the practical application of UV stabilizers.

7.1 Collaborative Effect of Multiple Stabilizers

Combination with Other Stabilizers

- * UV Absorbers: Combining HALS with UV absorbers, such as benzophenones or benzotriazoles, can provide a synergistic effect. UV absorbers prevent UV light from penetrating the polymer matrix, thereby reducing the formation of free radicals that HALS need to neutralize.
- * Antioxidants: Adding antioxidants, like hindered phenols, can help in scavenging peroxy radicals formed during photo-oxidation, complementing the action of HALS.

Experimental Evaluation

- * Synergy Studies: Can conduct experiments to evaluate the synergistic effects of combining HALS with other stabilizers. Measure the degradation rate, mechanical properties, and surface characteristics of PVC samples with various combinations of stabilizers.

7.2 UV Intensity Variation with Thickness

Application of Beer-Lambert Law

- * UV Absorption: As UV light penetrates the thickness of PVC, its intensity decreases according to the Beer-Lambert law:

$$I(x) = I_0 e^{-\alpha x}$$

where $I(x)$ is the UV intensity at depth x , I_0 is the initial UV intensity, and α is the absorption coefficient of PVC.

- * Depth Profile Analysis: Analyze the degradation profile across the thickness of PVC. Measure how the concentration of degradation products varies with depth to understand the protective effect of stabilizers throughout the material.

Simulation and Modeling

- * Depth-dependent Degradation: A simulation model can be developed that accounts for the depth-dependent UV intensity. That model can be used to predict the degradation rate and effectiveness of stabilizers at different depths within the PVC.

7.3 UV Intensity, Time, and Temperature Dependence

Dynamic UV Intensity

* Temporal Changes: Consider the changes in UV intensity over time, such as diurnal cycles and seasonal variations. Implement these variations in the kinetic models to predict degradation rates more accurately.

* Temperature Effects: Include the effect of temperature on the rate of photodegradation. Higher temperatures can increase the rate of chemical reactions, thereby accelerating degradation.

Modified Kinetic Equations

* Anton-Prinet et al. suggested Equation: Incorporate the modified kinetic equation proposed by Anton-Prinet et al. to account for the dynamic changes in UV intensity and temperature. The equation will help in better predicting the initiation rate constant under varying environmental conditions [27].

$$R_i = \alpha\Phi I + \alpha'k'_{I_0} \exp\left(-\frac{E_1}{RT}\right)$$

Where,

A - radicals formed from hydroperoxide by absorption of UV radiation,

α' - radical formed by thermolysis,

Φ - quantum yield of photolysis,

I - light intensity,

k'_{I_0} - constant of Arrhenius equation,

E_1 - apparent activation energy,

R - gas constant,

T - absolute temperature.

* Experimental Validation: Perform experiments to validate the modified kinetic model. Measure degradation rates under controlled UV and temperature conditions to compare with model predictions.

7.4 Other Possible Modifications

Surface Coatings

* Barrier Layers: Apply protective surface coatings that can further reduce UV penetration and enhance the effectiveness of internal stabilizers.

* Nanocomposites: Incorporate nanomaterials like TiO₂ or ZnO nanoparticles, which can act as UV blockers and provide additional reinforcement to the PVC matrix.

Advanced Characterization Techniques

* Real-time Monitoring: Use real-time spectroscopic techniques to monitor the degradation process and the effectiveness of stabilizers. Techniques like in-situ FTIR or Raman spectroscopy can provide valuable insights into the chemical changes occurring within the polymer.

Long-term Field Studies

* Environmental Exposure: Conduct long-term field studies to evaluate the performance of stabilized PVC under actual environmental conditions. This will help in understanding the real-world effectiveness of the proposed modifications.

CONCLUSION

This project successfully developed a comprehensive kinetic model that simulates the degradation of PVC under UV exposure. The model incorporates the relevant chemical reactions and rate equations, enabling the prediction of degradation rates and the evaluation of the effectiveness of Hindered Amine Light Stabilizers (HALS), particularly Tinuvin 770, in retarding this process.

The user-friendly software interface created alongside the model empowers users to predict the optimal concentration of Tinuvin 770 required for a specific application based on factors such as PVC formulation, UV intensity, and desired product lifetime. This translates to several key benefits:

Enhanced Durability of PVC Products: By optimizing PVC formulations with the appropriate stabilizer concentration, manufacturers can significantly improve the material's resistance to UV degradation, leading to longer product lifespans and reduced replacement costs.

Sustainable Practices: Extending the usable life of PVC products minimizes waste generation and the environmental impact associated with material production and disposal.

Market Competitiveness: Manufacturers equipped with this model can design PVC products with superior UV resistance, strengthening their market position and customer satisfaction.

Improved Regulatory Compliance: The model can aid in formulating PVC that adheres to environmental regulations concerning the release of hydrochloric acid during degradation.

Furthermore, the project has contributed to a deeper understanding of PVC degradation mechanisms through a detailed analysis of: Photochemical processes, including free radical formation, chain scission, and cross-linking.

The detrimental effects of UV degradation on various PVC properties, such as discoloration, loss of mechanical strength, and surface changes.

The stabilizing role of HALS in scavenging free radicals and interrupting the degradation cycle.

In conclusion, this project offers a valuable tool for the PVC industry. The kinetic model and software interface provide practical solutions for optimizing PVC formulations, ensuring product longevity, and promoting sustainable practices. The project's contribution to the understanding of PVC degradation mechanisms further paves the way for advancements in material science and responsible manufacturing.

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